Azure Data Engineering

ANSH LAMBA

service :-

IAAS (Infrastructure as a service) :-

* You have to manage all things.
* You manage applications, data, runtime, middleware & OS.
* Azure only manges services

PAAS (Platform as a service) :-

* You have to mange some staff only.
* You manage applications & data only. Ex:- DataLake
* Azure manges some things like runtime, middleware & OS.

SAAS (Software as a service) :-

* You no need to manage any things
* You just use the software. Ex:- email, outlook, microsoft 365
* Azure manages all things.

Azure Data service

* It is having tons of services.

1. Azure SQL DataBase :- (storage)

* It is used as a source
* It is a indirect service. * store structure Data

2. Data Lake :- (Storage)

* We have to do manage like downscale, upscale
* It is a direct service
* stores any kind of data structured, semi-structure & un structured.

3. Azure Cosmos db :- (Storage)

* It is ~~so~~ so sometimes You can mange it some time software engineers manager
* To store JSON Data.
* It is a indirect service.
* It is a source.

4. Events HUB:- (Stage)

* It is alternative to Apache Kafka
* It is useful for real-time data like IOT data.
* Pub/Sub model

5. Azure Data Factory (ADF):-

* It is a orchestrate / ETL purpose.

6. Databricks:- (Data processing)

* To transform Data.

7. Synapse Analytics:-

* It does everything orchestration, processing, data warging & SQL end point & more.
* It is a backbone of modern Data enginiring

8. Stream Analytics:-

* It was very rare
* It is used for real time streaming data.

AZURE DATA LAKE

Azure Data Lake is like a huge storage system where you can store any kind of data - structured (like tables), semi structured (like JSON, CSV), & unstructured (like images).

AZURE DATA FACTORY

ADF is microsoft Azure's cloud based data integration service - basically, it helps you move, manage & transform data from different sources in an automated way

Core Concepts:-

1. Linked services
2. Dataset
3. Activities
4. Pipelines.

AZURE DATA WAREHOUSE

* It is backbone for azure data enginiring.

Ex:- Synpase anyltics, AWS RedShift, Google bigQuery, snowflack, data bricks